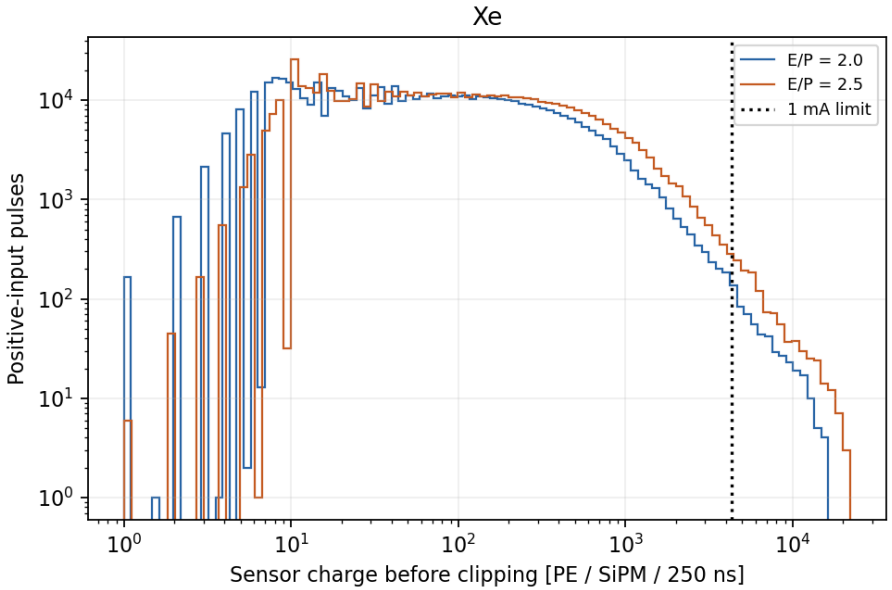
S14160-1315PS + TPB: E/P pilot

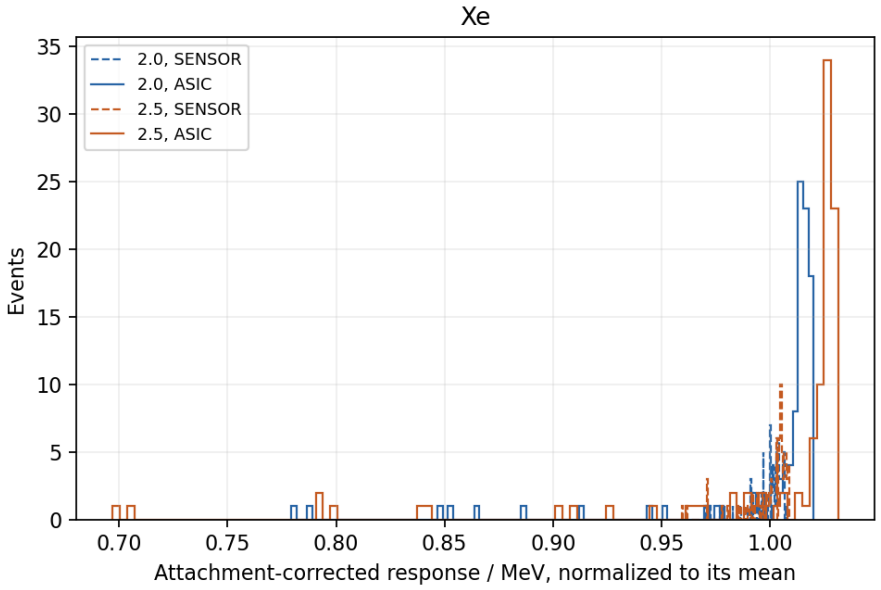
100 matched bb0nu events (IDs 0–99), 1 mm MC stepping, each gas and E/P point; 15 bar, 15% He. Centred 1.3 × 1.3 mm S14160-1315PS at zero gap, 5 mm hole diameter/depth, 10 mm pitch. Uniform isotropic axial EL emission; absorbing surroundings. TPB coats the sensor; complete VUV absorption, intrinsic conversion 0.62, forward fraction 0.5, blue PDE approximated by its 0.32 peak: effective PDE 0.0992. Prompt TPB stays in the parent bucket. 7,284 cells, gain 360,000, dark rate 120 kHz. Provisional recharge 12.5 ns with approved 25 ns sensitivity; uniform microcell illumination and constant PDE during recovery. ASIC: 1 mA, 250 ns, 3.5417 PE RMS (same input charge noise as the previous sensor). J = 0.030 at E/P = 2.0 and 0.015 at 2.5, for both gases. Nominal results use 1024 statistical substeps per 250 ns bucket.

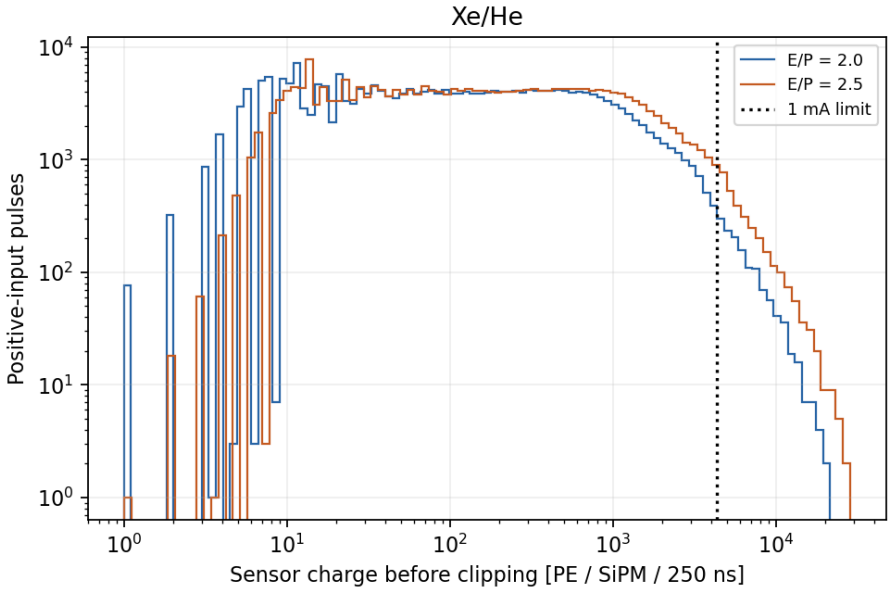
E/P	Gas	Mean detected PE [million]	Sensor / detected [%]	ASIC charge lost [%]	Events clipping [%]	Positive cells clipping [%]
2.0	Xe	0.933	99.43	1.284	25	0.095
2.0	Xe/He	0.895	98.70	3.367	58	0.591
2.5	Xe	1.301	99.13	2.276	30	0.215
2.5	Xe/He	1.268	98.15	6.248	97	1.423

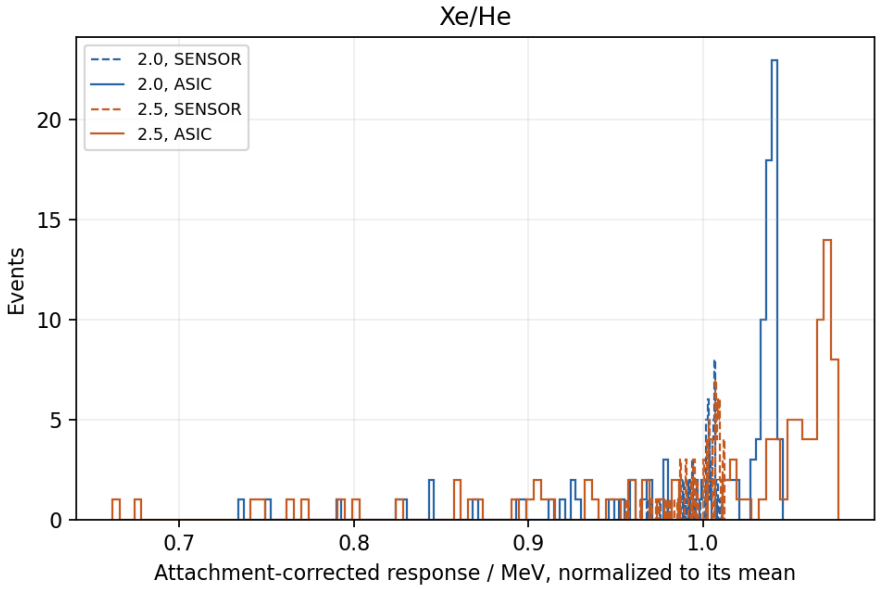
Widths are full-distribution RMS/mean. Known attachment is corrected event by event and totals are divided by the deposited MC energy; sensor/ASIC sums have expected stationary dark baseline subtracted. Recovery and clipping remain in the forward response. The detected-PE width isolates optical counting statistics; the sensor and ASIC widths expose nonlinear response. These are not resolutions after a fitted nonlinearity correction. The raw total distributions also include energy deposition and drift variation. Each pulse is one SiPM–250 ns cell. The logarithmic pulse plot displays positive sensor output; zero outputs remain in all event sums and pulse means.

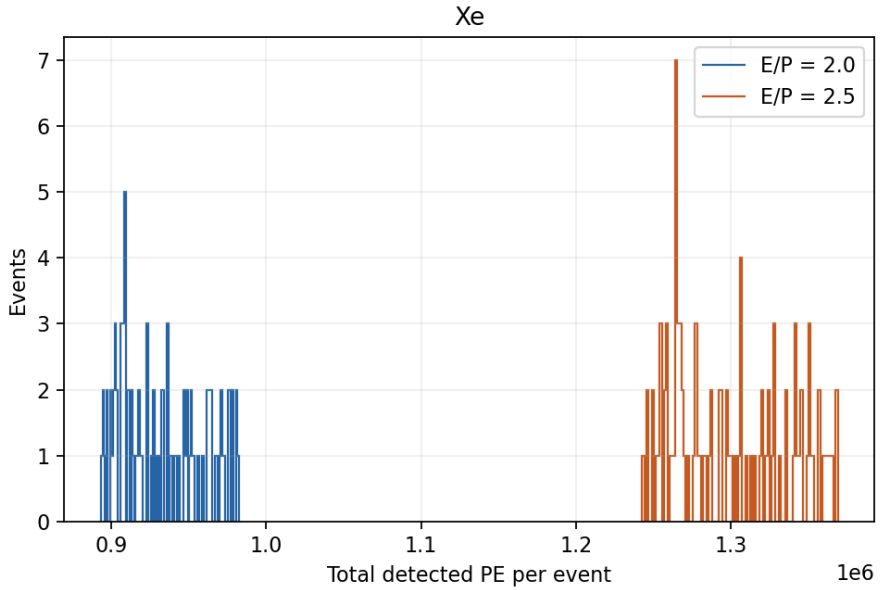
E/P	Gas	VUV RMS [%]	Detected PE RMS [%]	Sensor RMS [%]	ASIC RMS [%]
2.0	Xe	0.141	0.170	0.732	4.520
2.0	Xe/He	0.139	0.175	0.877	6.784
2.5	Xe	0.139	0.165	0.986	6.645
2.5	Xe/He	0.141	0.169	1.212	9.804

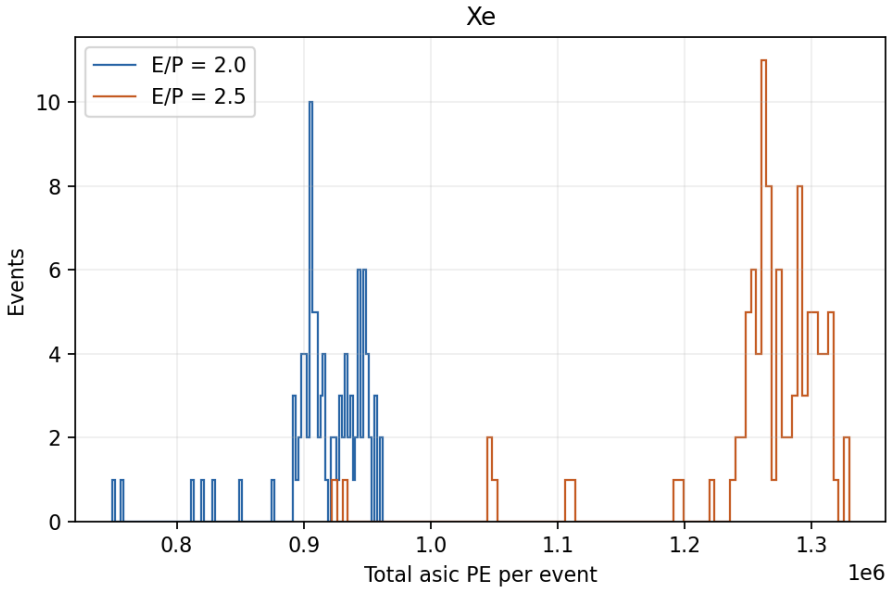

Xe
E/P = 2.0
E/P = 2.5
1 mA limit
Positive-input pulses
Sensor charge before clipping [PE / SiPM / 250 ns]

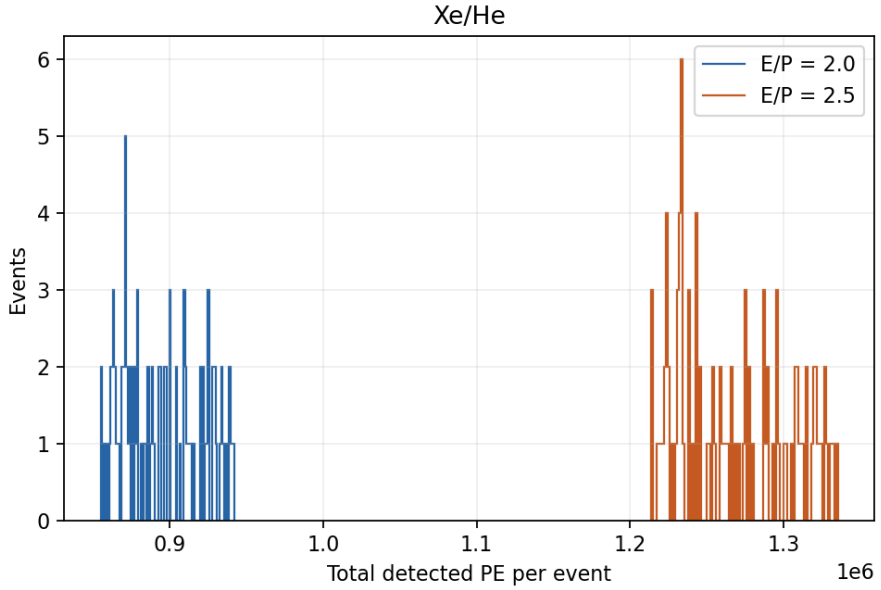

Xe
2.0, SENSOR
2.0, ASIC
2.5, SENSOR
2.5, ASIC
Events
Attachment-corrected response / MeV, normalized to its mean

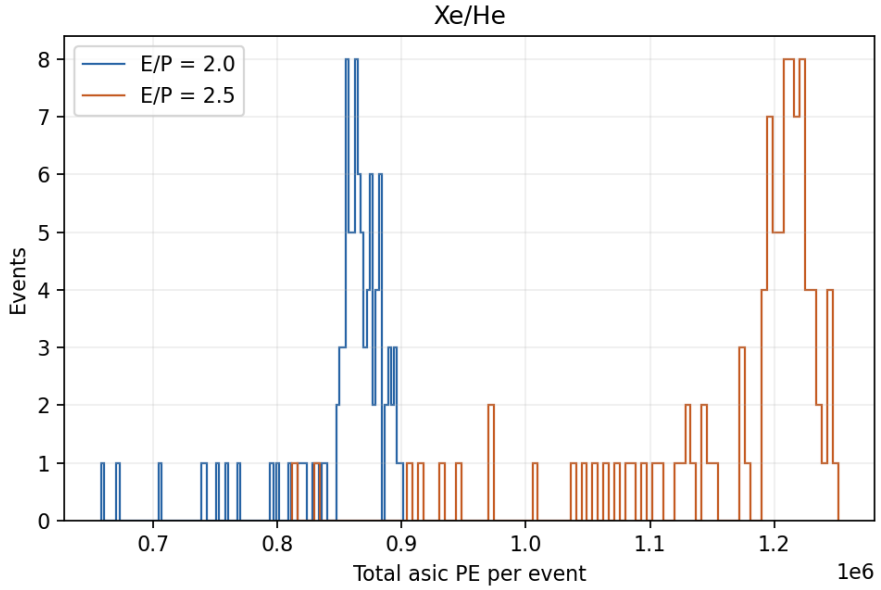

Xe/He
E/P = 2.0
E/P = 2.5
1 mA limit
Positive-input pulses
Sensor charge before clipping [PE / SiPM / 250 ns]


Xe/He
2.0, SENSOR
2.0, ASIC
2.5, SENSOR
2.5, ASIC
Events
Attachment-corrected response / MeV, normalized to its mean


Xe
E/P = 2.0
E/P = 2.5
Events
Total detected PE per event
1e6


Xe
E/P = 2.0
E/P = 2.5
Events
Total asic PE per event
1e6


Xe/He
E/P = 2.0
E/P = 2.5
Events
Total detected PE per event
1e6


Xe/He
E/P = 2.0
E/P = 2.5
Events
Total asic PE per event
1e6

Pulse and sensitivity summaries

E/P	Gas	Event-mean PE: mean	Event-mean PE: median	Event-max PE: mean	Event-max PE: median
2.0	Xe	250.8	155.0	3412.9	2054.6
2.0	Xe/He	493.5	351.9	6241.6	4562.5
2.5	Xe	347.8	215.5	4679.8	2842.0
2.5	Xe/He	693.6	495.3	8640.5	6346.1

Recharge ns / steps	E/P	Gas	ASIC loss [%]	Sensor RMS [%]	ASIC RMS [%]
12.5 / 512	2.0	Xe	1.282	0.719	4.503
12.5 / 1024	2.0	Xe	1.284	0.732	4.520
25.0 / 512	2.0	Xe	1.116	1.372	4.663
12.5 / 512	2.0	Xe/He	3.365	0.878	6.775
12.5 / 1024	2.0	Xe/He	3.367	0.877	6.784
25.0 / 512	2.0	Xe/He	2.932	1.670	6.853
12.5 / 512	2.5	Xe	2.272	0.998	6.650
12.5 / 1024	2.5	Xe	2.276	0.986	6.645
25.0 / 512	2.5	Xe	1.973	1.855	6.779
12.5 / 512	2.5	Xe/He	6.242	1.220	9.806
12.5 / 1024	2.5	Xe/He	6.248	1.212	9.804
25.0 / 512	2.5	Xe/He	5.449	2.249	9.826

E/P	Gas	Mean sensor change [%]	Clip loss change [percentage points]
2.0	Xe	+0.0036	+0.0012
2.0	Xe/He	-0.0012	+0.0016
2.5	Xe	-0.0107	+0.0039
2.5	Xe/He	-0.0090	+0.0057

Physical references

- Benson et al. (2018), intrinsic TPB QE, Figure 19
- Hamamatsu S14160 datasheet
- Klest and Zurek, measured pulse decay, slide 16